

PHILOSOPHY OF SCIENCE

Lecture 6: Replies to the PMI
Enrico Cinti
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OUTLINE

- (A) Summary of arguments about scientific realism
- (B) Recent discussions
- (C) Extensional scientific realism

(A) SCIENTIFIC REALISM

A summary of arguments so far



SCIENTIFIC REALISM

- One of the **central debates** in current philosophy of science.
 - It deals with what is widely taken to be one of the central aims of science: to provide knowledge.
- Contemporary debates about scientific realism are in part reactions to the problems of **induction and under-determination**.
 - Not only! Other lively debates about specific forms of scientific realism.
- Also touches on other important topics, such as **explanation**.
- **Today:** further development of the debate
 - Especially Stanford's historicist criticism
 - (A sketch of) SDH's reply to Stanford's critique

SUMMARY OF THE ARGUMENTS: FOR REALISM

- **Inference to the best explanation.** *Some force.*
 - Van Fraassen: no such rule. Empirical adequacy: competing hypothesis
- **No-miracles argument.** *Some force.*
 - Prediction of novel facts/regularities
 - Van Fraassen's Darwinism
 - Musgrave: Darwinian/historical vs. intensional (conceptual) explanation
- **The demand for explanations.** *Some force.*
 - Explanation vs. ultimate explanation
 - Adequate explanations requiring truth
- **No theory-observation distinction.** *Some force.*
 - Stances? Voluntarism?

SUMMARY OF ANTI-REALIST ARGUMENTS

- **Empirical under-determination:** under-determination of theory by the empirical data. *Stalemate/not much force?*
 - Virtues (simplicity), pragmatics, metaphysics
- **Constructive empiricism:** *main contender*
 - Not all problems of logical empiricism, some of the virtues of scientific realism
 - 'Observable' remains a problem.
- **Pessimistic meta-induction (Laudan):** *Some force.*
 1. No good notion of 'approximate truth'
 2. No connection between empirical success and approximate truth
 3. Reference required for truth, but old theories contain obviously non-referring terms: belief in the truth of theories not justified

EMPIRICAL UNDERDETERMINATION

- It seems possible that the same evidence might confirm different theories
- **Transient vs. permanent underdetermination:** can new evidence break the underdetermination?
- Difficulties in identifying realistic examples of underdetermination:
 - It's easy to build logically inequivalent theories, but these are toy-models
 - Real physics examples are exceedingly rare and controversial
- **Realist reply:** theoretical virtues break the underdetermination
- Simplicity, beauty, fruitfulness: non-empirical, but still truth-conducive
- Empirically equivalent theories can still disagree on their theoretical virtues: underdetermination broken

THE PESSIMISTIC META-INDUCTION ARGUMENT (PMI)

- **History of science:** many theories that we now consider false were once empirically successful
- These theories assumed that terms like 'aether', 'phlogiston', etc. referred
- But we do not now take these terms to refer
 - Theories assume entities that we now consider non-existent
- **Induction:** presently successful theories will (may well) turn out to be false too
 - Not justified in believing that theories are true
- No reason to assume that the entities, properties, and processes postulated by our **best theories** are real

LAUDAN'S EXAMPLES

crystalline spheres of ancient and medieval astronomy

humoral theory of medicine

effluvial theory of static electricity

catastrophist geology, with its commitment to a universal

~~deluge~~
phlogiston theory of chemistry

caloric theory of heat

vibratory theory of heat

vital force theories of physiology

electromagnetic aether

optical aether

theory of circular inertia

theories of spontaneous generation

'This list, which could be extended *ad nauseam*, involves in every case a theory which was once successful and well confirmed, but which contained central terms which (we now believe) were non-referring.'

(B) RECENT DISCUSSIONS

Selective confirmation, structural realism and Stanford's critique





RECENT DISCUSSIONS

- Debate about scientific realism is ongoing.
- Kitcher (1993) and Psillos (1999) point out that van Fraassen's notion of realism is incorrect: realists do not believe in the **literal truth** of theories, but in their *approximate or probable truth*.
 - 'Science aims to give us, in its theories, a **literally true** story of what the world is like; and **acceptance** of a scientific theory involves the **belief** that it is true.'
- Scientific theories are not monolithic entities, but have "parts", some of which are more important than others. Their accounts aim to make precise the notion of 'approximate truth'.

REPLIES TO THE PESSIMISTIC META-INDUCTION (AND COUNTER-REPLIES)

- **Mature theories only:**

- ✓ Boyd (1981), Hardin and Rosenberg (1981): be a realist only about **empirically well-confirmed**/mature theories
- Stanford (2006): the requirement of 'maturity' is **arbitrary** and post-hoc
- Surely some of the theories in Laudan's list were well-confirmed? Phlogiston theory of combustion, caloric theory of heat

- **Selective confirmation:**

- ✓ Kitcher (1993): better analysis of 'reference', so that not all terms of theories refer
- ✓ Psillos (1999): *divide et impera*. Different theory parts confirmed differently

- **Structural realism:** continuity is given by preserved mathematical structure

- Not enough to specify a (scientific) ontology?

KITCHER VS. STANFORD ON REFERENCE

- **Kitcher (1993):**

- Terms used by scientists **do not automatically refer**
- Look at specific context to see what terms refer to
- 'Aether' could refer to the electromagnetic field in some cases, to empty space in others, and to nothing at all in others.
- **Reference is fixed differently in different cases**
- Laudan wrong to conclude that these terms are non-referring

- **Stanford (2006):**

- Agrees some such distinction might make sense. But the argument misses the point of meta-induction:
- Even if central terms refer, they are embedded in **theories that repeatedly turn out to be radically misguided**. Such theories cannot be approximately true

STANFORD ON REFERENCE

Kitcher's explanation 'little comfort to the realist if we insist that some tokens of terms like 'dephlogisticated air' in rejected theories referred after all, while admitting that the relevant theoretical accounts and descriptions of those entities were **mistaken about virtually everything except the fact that the entities in question played some causal role** in producing observable phenomena'.

PSILLOS' (1999) SELECTIVE CONFIRMATION

- *Divide et impera*. Distinguish different **parts of theories**:
 - **'Causal core'**: role in explaining the phenomena, responsible for the theory's success. Causal core typically retained by later theories
 - **'Idle constituents'**: not causally involved in explaining phenomena, not responsible for success. These constituents typically disappear in later theories.
 - Compare with Lakatos (hard core/protective belt)!
- Example: Maxwell's theory and the aether
 - Causal core: theory's actual equations/mechanisms of explanation, which *do not require the aether*
 - Aether is idle constituent, not retained in modern electromagnetic theory
- Mixed theories of reference (Papineau, Lewis)
 - Combine 'causal' and 'descriptive' accounts of reference



STANFORD ON SELECTIVE CONFIRMATION

- Distinction between causal core and idle terms is **anachronistic**, would have been denied by the scientists who developed those theories
 - Maxwell regarded the aether as essential part of electromagnetic theory
 - How can we say that Maxwell's theory "was really not about the aether"?
- **Virtually guaranteed** convergence towards causal core, which appears in current theories
 - Almost by definition, whatever entities retained by current theories are part of what we now call the causal core
 - Causal core need not be what scientists of the past thought was important for their theories

STANFORD ON SELECTIVE CONFIRMATION

- Need **prospectively applicable criteria** of idleness and/or selective confirmation: avoid Whig history
 - Scientists of the future will make the same verdicts about our current theories
 - Without arguments that treat history in a way that is not anachronistic, not entitled to be realists about our best theories
- Selective confirmation requires either knowing facts about the beliefs and intentions of past scientists, or require us to say that past scientists **repeatedly misidentified the parts/features/aspects** of theories that contributed success
 - 'Pyrrhic victory' for scientific realism!

(C) EXTENSIONAL SCIENTIFIC REALISM

A reply to the PMI and PUA



WAY FORWARD: AGREEING WITH STANFORD

- Stanford's (2006) critique of selective confirmation seems generally on the mark:
1. **Need to explain the connection between *reference* and *approximate truth***
 - ✓ Solution needs to explicate both
 2. **Avoid projecting current insights into the past:** parts of past theories that now seem true, or what we now believe the theories "are really about", on past scientists' intentions and beliefs: these are **incorrect readings of history**.
 - ✓ Methodological principle: read the arguments that past scientists used to explain phenomena **in their historical context**.

CRITICISING STANFORD

- Stanford's claim that convergence towards a causal core shared between old and new theories is "virtually guaranteed" seems incorrect
 - If core is obtained by eliminating idle terms while retaining the ones that cause the phenomena, it could well be that **none of the original terms remain**. No guarantee that causal core remains.
 - Any *causal* core that remains is highly remarkable.
- Stanford **reads into Maxwell's text** when he concludes that Maxwell considered the idea that electromagnetic could be transmitted in a different way than through an aether, "incoherent" and "unintelligible".
 - Stanford's critique of continuity in selective realism not strong.

THE PERPLEXITY ABOUT REFERENCE

- Weakness in the literature, from Putnam (1978) to Stanford (2006): assumption of **a single kind of meaning of words**
- Hence **Laudan's perplexity** about terms of old theories "failing to refer" ('aether', 'phlogiston', etc.)
- There are plausible arguments that suggest both that 'aether' refers, and that it does not refer
- Indicative of the need for a further linguistic/conceptual distinction

LAUDAN'S EXAMPLES

crystalline spheres of ancient and medieval astronomy

humoral theory of medicine

effluvial theory of static electricity

catastrophist geology, with its commitment to a universal deluge

phlogiston theory of chemistry

caloric theory of heat

vibratory theory of heat

vital force theories of physiology

electromagnetic aether

optical aether

theory of circular inertia

theories of spontaneous generation

**Claim: not at all
'obvious' that the
terms in these theories
"fail to refer"**

EXTENSIONAL SCIENTIFIC REALISM

- **Cautious realism**, as against van Fraassen's characterization:
 - Belief in **literal truth** of theories is often naïve
 - Not "truth of theories, period", but **approximate truth**
- First aim: give cogent answer to pessimistic meta-induction argument, as the main argument open against scientific realism
- Idea: problem of pessimistic meta-induction argument is that it considers a single notion of meaning
 - Two types of meaning in philosophy of language (Frege, Carnap, Lewis)
- Related: Hacking (1983), Radder (1985), Galison (1997), Massimi (2016)

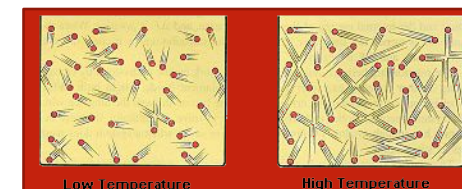
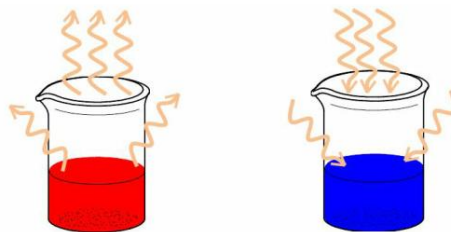
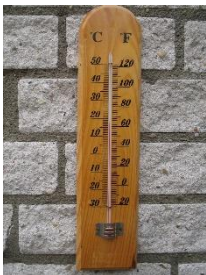


SENSE AND REFERENCE/ INTENSIONS AND EXTENSIONS

- Venus is one of the brightest objects in the sky. Seen both morning and evening, called both 'morning star' and 'evening star'.
- Different linguistic meanings:
 - **Morning star:** brightest star in the morning
 - **Evening star:** brightest star in the evening
- Still, in our planet both meanings refer to a single object: planet Venus.
- Call this distinction the 'sense' and the 'reference' (also: 'intension' and 'extension' of the words: Carnap's jargon).
 - **Sense (= intension):** linguistic meaning of the words. *Morning/evening star*
 - **Reference (= extension):** actual object/entity the words refers to. *Venus*

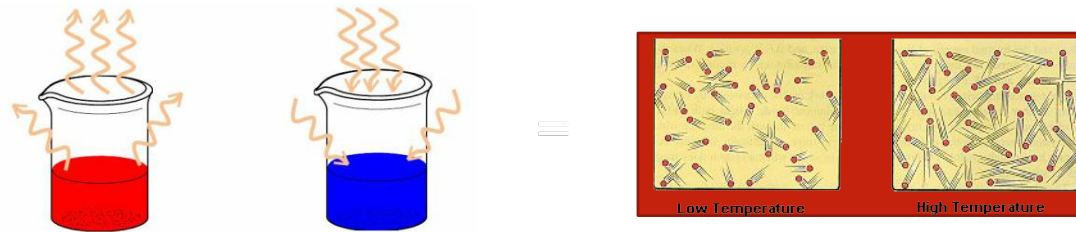
THREE NOTIONS OF TEMPERATURE

- Distinction between **intension** and **extension** important for scientific theories, because we are interested in how words refer.
- Consider different uses of '**temperature**':
 - Lay person uses 'temperature' without necessarily being able to give a scientific explanation of it. Would say something like: 'temperature is the expansion of mercury that I measure with a thermometer'.
 - Physicist trained in **thermodynamics**: 'temperature is a measure of the tendency of an object to give energy to its surroundings'.
 - Physicist trained in **statistical mechanics**: 'temperature is the mean kinetic energy of the molecules in a substance'



INTENSIONS AND EXTENSIONS

- **Different meanings that scientists give to same word** can be perplexing
 - Kuhn: scientists 'live in different worlds'
- One way to understand this: word 'temperature' has **different intensions (senses) but the same extension (reference)**



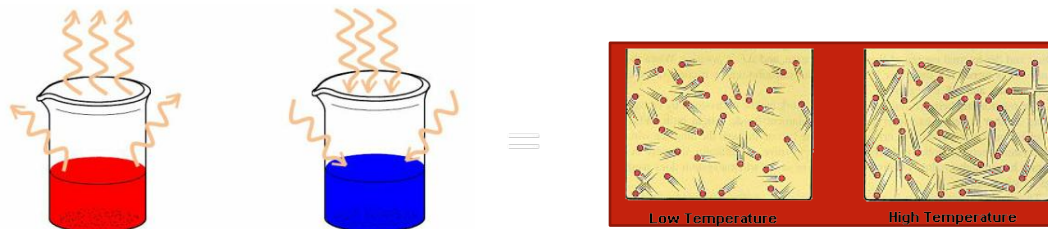
- Two physicists give different intensions to 'temperature', because their definitions are different
- However, the different intensions refer to the same physical quantity (the two words 'temperature' have the same extension)

ANSWERING THE PESSIMISTIC META-INDUCTION

- **Need to secure continuity of reference and approximate truth between discarded-new theories**
- **Step 1:** continuity in extensions of theories that follow one another:
extensional equivalence reference to same items in **domain of application**
 - Extensional equivalence quantum/classical mechanics : can derive all of the extensional results of classical mechanics from quantum mechanics
 - Contrast with Psillos: continuity here not secured by “theory parts”, but by delimiting the theory’s extension to a domain of application
- **Step 2:** define ‘approximate truth’ in terms of extensions
- Idea/slogan: *We are justified in being scientific realists about extensions but not necessarily about intensions.*

EXTENSIONAL EQUIVALENCE AND CORRESPONDENCE

- Temperature (hot/cold) and temperature (fast/slow) are *extensionally equivalent concepts*
- What is the extension of a term in a scientific theory? Recall Venus: extensionally, i.e. in our world, *Morning star* $\equiv$ *Evening star*; so extensionality is our world/conditions of observation
- **Extension is determined by the intension and by the specific circumstances/context where a given phenomenon is studied**



EXTENSIONS

- **Aspects that determine extensions in natural science:**
 - **Specification of a model of some phenomenon:** including choices of initial/boundary conditions, assumptions about the population, environmental conditions, etc.
 - **Values of the theory's and model's free parameters:** mass, coupling strength, chemical concentration,...
 - **Extra-theoretical facts not captured by the theory or model:** experimental errors, other influences, material realisation, weather conditions,...
 - **Aproximations and idealizations:** taking a limit $v \ll c$ (special relativity), $\hbar \rightarrow 0$ (quantum mechanics), $G \rightarrow 0$ (general relativity).
- Related view: **effective realism** in high-energy physics.
 - Realism about the theory **in a specified range of parameters.**

EXTENSIONAL EQUIVALENCE

- How do we establish that the extensions of (the terms of) different theories are equivalent?
- Three aspects:
 1. **Predictive-theoretical (numerical and formal) correspondence:** the two theories make the same predictions, under the relevant approximations.
 2. **Material correspondence** (including experimental, instrumental correspondence and replicability): the system studied is the same, but we study it from different points of view.
 3. **Conceptual correspondence:** the concepts of the two theories are not identical, but “match” (play the same roles) in the given extension.

EXAMPLES

- **Predictive correspondence:** quantum mechanics reproduces the results of classical mechanics, under specific conditions (large systems etc.).

$$\langle dp/dt \rangle = -\langle \nabla V \rangle \rightarrow F = -\nabla V \quad (\text{Newton's equation})$$

- **Material correspondence:** the atom studied by (i) the quantum theorist, (ii) the inorganic chemist is the same entity, even if their experiments are different (particle accelerator vs. electron microscope vs. chemical reaction).

- **Conceptual correspondence:** quantum and classical mechanics both contain the notions of 'position of a particle', play the same roles on a given extension.

THE AETHER, REVISITED

- Main argument of the pessimistic meta-inductivists:
- **Aether does not refer** (does not exist at all!), so how can we say that the old electromagnetic theory was true?
- How can it even be 'approximately' true, if it relies on the postulation of non-existing entities (Stanford: its ontology is "radically misguided")
 - Old and new electromagnetic theories are in *predictive* and *material* correspondence, but no *conceptual* correspondence
- Answer: '**aether**' **does refer**—the aether does exist.

THE AETHER, REVISITED

- What Maxwell called the 'aether' is extensionally equivalent to what we now call 'the electromagnetic field', together with a fixed frame of reference, $v \ll c$.
- Maxwell's argument for the existence of the aether was very good: electromagnetic energy does not disappear from the sender to pop up at the receiver at a different point in space (conservation of energy).
- The 'medium' that Maxwell is talking about is the same as the **electromagnetic field, together with a fixed frame of reference**.
 - 'Aether' and 'electromagnetic field' otherwise of course intensionally distinct!

A PYRRHIC VICTORY?

- Selective confirmation requires knowing facts about the beliefs/intentions of past scientists, or to say that past scientists repeatedly misidentified parts of theories that contributed to success.
- Extensional scientific realism does not distinguish theory “parts”
 - **Linguistic/philosophical theory about how to interpret scientific statements**
 - For all we know (assuming empirically adequate theories/sound scientific arguments), all the terms that scientists claim refer, do refer
- Compare van Fraassen on interpreting a theory literally, but believing only what it says about the observable
- Likewise, extensional scientific realism: **take literally what the scientist says, but only believe it as extensional not intensional kind of meaning**

APPROXIMATE TRUTH (A SKETCH)

- Use Laudan's idea of the progress of scientific theories, measured by the relative number and significance of the problems these theories solve
- Use this to define 'approximate truth':
- T is closer to the truth than T' iff it is extensionally true in a larger, and more significant, domain of application
- Is there a danger that theories will become more vague as they become more general? Standard answer from philosophy of science: T should be as good as T' as a theory, e.g. concerning predictive power and explanation

COMPARING TWO THEORIES

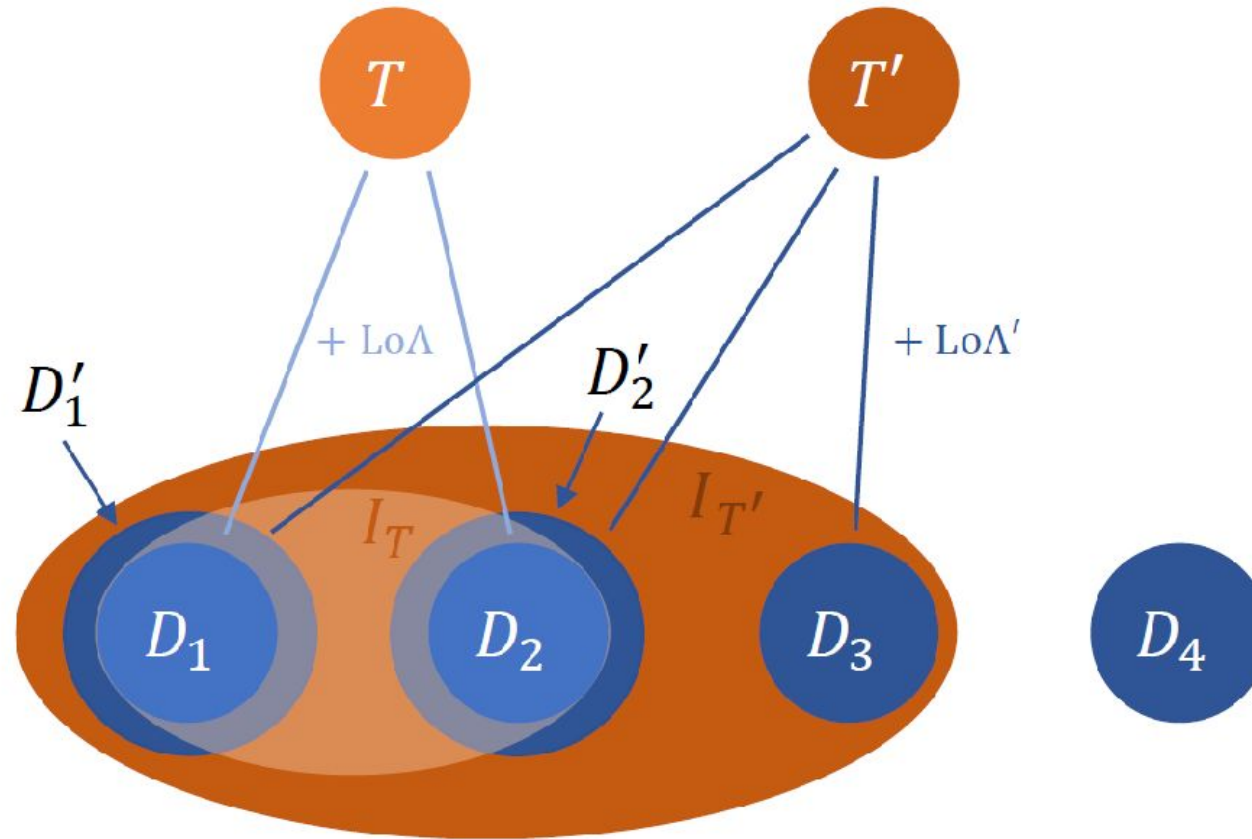


Figure 3: Extensional truths of the theories, T and T' , over the various domains (light and dark blue lines). T 's intension is $I_T = D_1 \cup D_2$, and T' 's intension is $I_{T'} = D'_1 \cup D'_2 \cup D_3$. The domain in which T' is true is larger than that of T , and $D_1 \cup D_2 \subset D'_1 \cup D'_2$.

CONCLUSION

- Perplexity about reference originates in **too simple/naïve readings of history of science**
 - Glosses over important aspects of semantics: distinction intensions/extensions
- While Stanford's criticism partly on the mark, his insistence on "broad pattern(s) of repeated, profound, and unpredictable changes in fundamental theoretical orthodoxy" misses the existence of **restricted domains of application** where there is extensional equivalence
 - Continuity of reference: aether example, quantum mechanics
 - Conceptual, material, and predictive correspondence of theories
 - A (beginning of) a notion of approximate truth

REFERENCES

- Laudan, *A Confutation of Convergent Realism* (Curd & Cover)
- Kitcher, 1993, *The advancement of science: Science without legend, objectivity without illusions*
- Stanford, 2006, *Exceeding our grasp: Science, history, and the problem of unconceived alternatives*
- SDH, *An Extensional Scientific Realism (I)*, on canvas
- SEP entry Theory and Observation in Science:
 - <https://plato.stanford.edu/entries/science-theory-observation>
- SEP entry on Theoretical Terms in Science:
 - <https://plato.stanford.edu/entries/theoretical-terms-science>